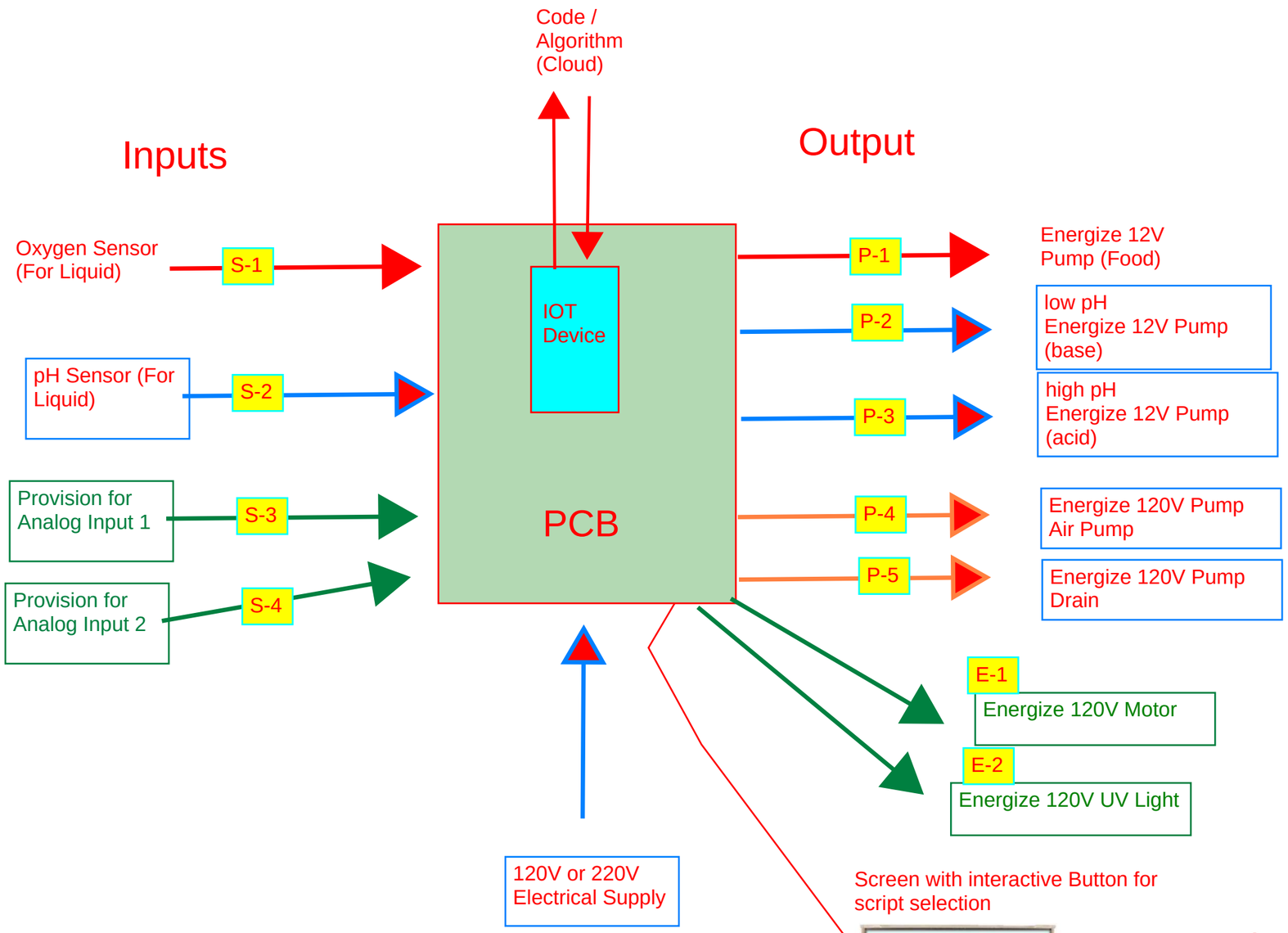
LogN
Basis of Design

Schematic



Screen with interactive Button for script selection





Screen showing
a) Script Selected
b) Duration
c) Analog Conditions (pH)
d) Analog Condition (Oxygen Content).

SCRIPT

Operator Select Script

Parameters/Variables monitored for duration

Parameters
- a) Sterilization Duration: X
- b) Duration: Y
- c) Input Frequency: Z
- d) Oxygen: O
- e) pH: P

Frequency: Input Analyzed every Z minutes and pumps are energized based on input

S-1 (Oxygen)

Parameter S-1 is evaluated.

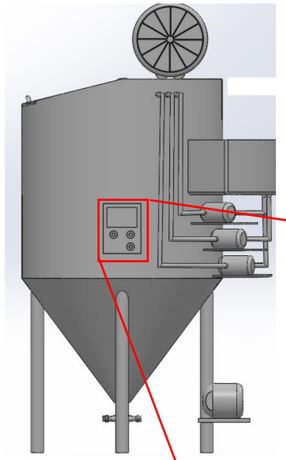
IF S-1 > O, Energize P-1

S-2 (pH)

Parameter S-2 is evaluated.

IF S-2 > P, Energize P-3
IF S-2 < P, Energize P-2
IF S-2 = P, Nothing

More information on Page 3: Firmware Explanation



Sensor Housing



Energize 120V Motor E-1

Energize 120V UV Light E-2

On/Off

Start/Stop

Screen Display

P-4 → Energize 120V Pump

P-5 → Energize 120V Pump

Oxygen Sensor (For Liquid)

pH Sensor (For Liquid)

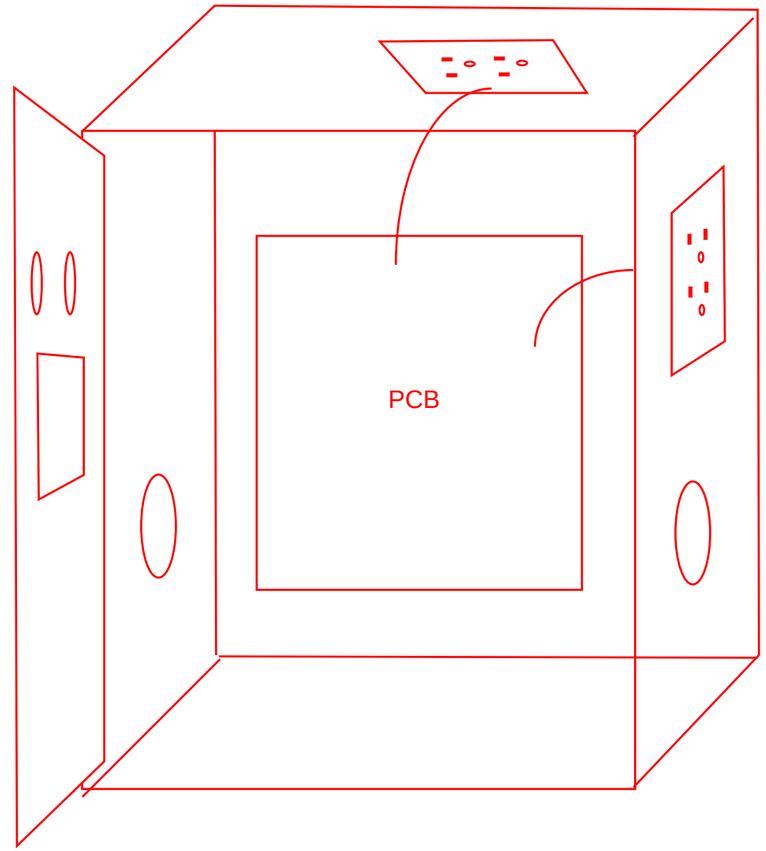
Energize 12V Pump

low pH Energize 12V Pump

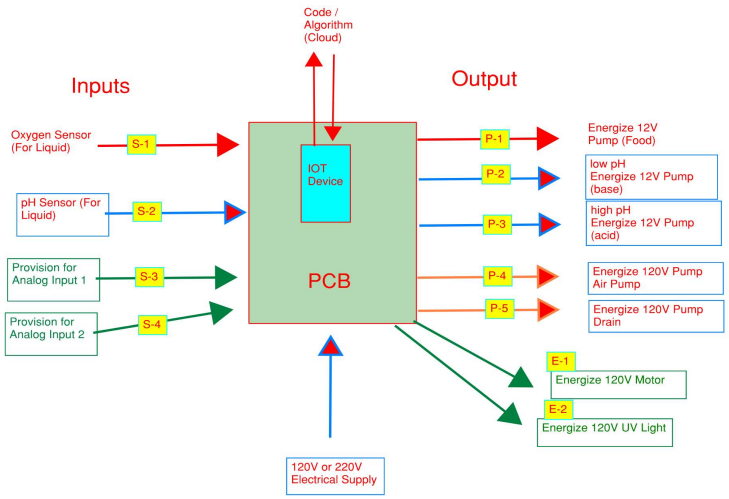
high pH Energize 12V Pump

Power Cord for Controller





Firmware Explanation



CODE BASE											
Parameters/Variables >	Duration	Input Frequency	Oxygen	pH	ENERGIZE						
Script Name	Y	Z	O	P	P-1	P-2	P-3	P-4	P-5	E-1	E-2
Microbe 1					S-1 > O	S-2 < P	S-2 > P	ON	ON	ON	OFF
Microbe 2					S-1 > O	S-2 < P	S-2 > P	ON	ON	ON	OFF
Microbe 3					S-1 > O	S-2 < P	S-2 > P	ON	ON	ON	OFF
CT-1					S-1 > O	S-2 < P	S-2 > P	ON	ON	ON	OFF
CT-2					S-1 > O	S-2 < P	S-2 > P	ON	ON	ON	OFF
Sterilization	n/a	n/a	n/a	n/a	n/a	n/a	n/a	OFF	OFF	ON	ON

PERFORMANCE RUN

Operator Engagement — Code Base





1 Operator Selects Script interacting with the Controller. (Simple Programmable Screen with Button Interface)

2



3 Operator Selects '**RUN**' (Button Interface)

4

5 Script is followed per Parameters/Variables

6

7 Script Conditions >

PARAMETERS / VARIABLES				ENERGIZE						
Duration	Input Frequency	Oxygen	pH	P-1	P-2	P-3	P-4	P-5	E-1	E-2
Y	Z	O	P	S-1 > O	S-2 < P	S-2 > P	ON/OFF	ON/OFF	ON/OFF	ON/OFF

8

9 Script is followed for Duration Y

10

11 P-4 Energized ON/OFF

12 P-5 Energized ON/OFF

13 E-1 Energized ON/OFF

14 E-2 Energized ON/OFF

15

16 Input Frequency Z

17 Every 'Z' Minutes Run the Following

18

19

20

21 Y

22

23 Z

24

25

26

27

28

29

30

31

32

33

34

35 Script Complete

36

Input S-1 = S-1
Input S-2 = S-2

		ENERGIZE	Duration
IF	S-1 > O	P-1	5s
IF	S-2 < P	P-2	5s
IF	S-2 > P	P-3	5s



37 Operator Select '**END**' to clear cache